Mawlana Bhashani Science and Technology University
Department of Information and Communication Technology
3rd Year, 2nd Semester Project

WIRELESS AUDIO STREAMER

PC to Mobile High-Fidelity Audio Streaming via Bluetooth PAN



COMPONENTS

- WASAPI Loopback Capture (SoundCard API)
- Python 3.12 Asynchronous Engine (aiohttp)
- Full-Duplex WebSocket Protocol Server
- Bluetooth PAN Adapter (BNEP TCP/IP)
- HTML5 Web Audio API / AudioWorklet Client
- Desktop Tkinter GUI with VU Level Meter

SKILLS

- Python & Asyncio Concurrency
- Windows Core Audio (WASAPI)
- Web Audio API & AudioContext
- WebSocket Real-Time Streaming
- Bluetooth PAN & Ad-hoc Networking
- Digital Signal Processing (PCM / VU)

ADVANTAGE

- Zero App Installation on Mobile (Web PWA)
- No Wi-Fi Router or Internet Required
- Uncompressed 48kHz High-Fidelity Sound
- Ultra-Low Latency (~50ms Adaptive Buffer)
- Real-Time VU Level Meter & Dynamic QR
- Universal Mobile Compatibility (Android/iOS)

SUPERVISED BY

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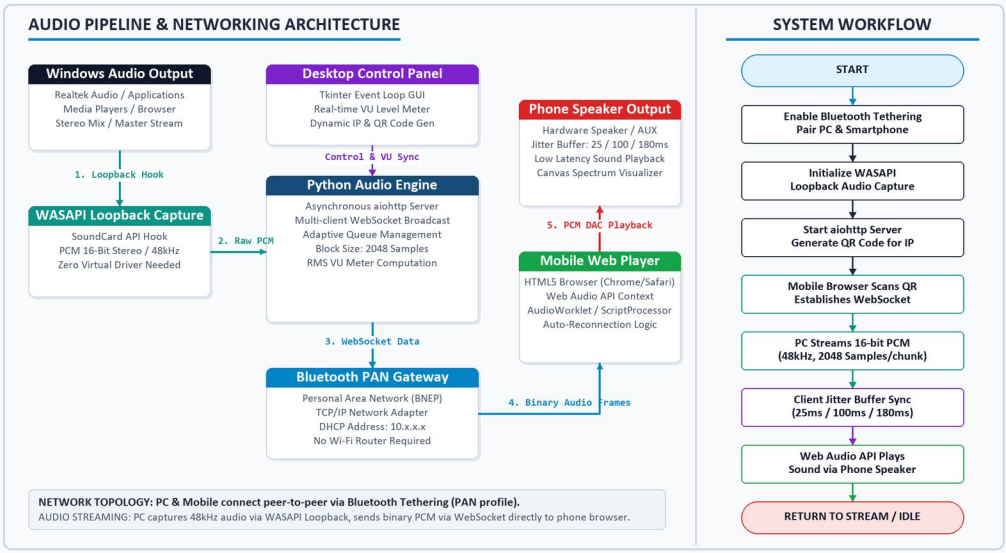
INTRODUCTION

This project presents an innovative wireless audio streaming system that converts any smartphone into a real-time external speaker for a Windows PC using only Bluetooth Tethering (PAN). By utilizing WASAPI loopback capture, the system captures master PC audio without third-party virtual audio cables, transmits low-latency 16-bit PCM chunks over WebSockets via an ad-hoc TCP/IP link, and renders high-fidelity sound through the phone's native browser via the Web Audio API—requiring zero app installation on the client device.

PROJECT OBJECTIVES

- **Driverless Audio Capture:** Hooks into Windows Core Audio via WASAPI loopback to record master audio in real time with zero driver installation.
- **Standalone Bluetooth Link:** Establishes a private TCP/IP data channel over Bluetooth PAN (Personal Area Network) without requiring a Wi-Fi router.
- **Universal Zero-Install Client:** Streams raw PCM binary audio to an HTML5 Web Audio browser receiver, supporting both Android and iOS out-of-the-box.
- **Adaptive Latency Management:** Features configurable jitter buffers (25ms / 100ms / 180ms) to ensure smooth playback without stuttering.
- **Desktop GUI & QR Pairing:** Equipped with an intuitive Tkinter control dashboard featuring a live VU meter and auto-pairing QR code.

WORKING PROCEDURES



CONCLUSION

Powered by Python 3.12, WASAPI loopback capture, and modern Web Audio standards, this project delivers a reliable, zero-cost wireless audio solution that transforms any smartphone into a high-performance wireless PC speaker over Bluetooth with minimal latency and crystal-clear acoustic output.

Presented by

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